

UNIT 1: Cryptography Basics

Introduction

Cryptography secures communication.

Detailed Explanation

Concepts

Encryption and decryption.

Goals

Confidentiality, integrity, authentication.

Applications

Used in secure communication.

Summary

This unit explains cryptography basics.

UNIT 2: Symmetric Key Cryptography

Introduction

Uses a single key.

Detailed Explanation

Algorithms

AES, DES.

Applications

Used in secure systems.

Summary

This unit explains symmetric crypto.

UNIT 3: Asymmetric Cryptography

Introduction

Uses public and private keys.

Detailed Explanation

Algorithms

RSA.

Applications

Used in digital signatures.

Summary

This unit explains asymmetric crypto.

UNIT 4: Network Security Protocols

Introduction

Ensure secure communication.

Detailed Explanation

Protocols

SSL/TLS.

Applications

Used in internet security.

Summary

This unit explains protocols.

UNIT 5: Cyber Security Practices

Introduction

Protect systems.

Detailed Explanation

Practices

Firewalls, antivirus.

Applications

Used in organizations.

Summary

This unit explains security practices.